📘 EduTrack

Academic Management Platform — Work Breakdown Structure

👥 Team 15 — VTU Internship 2026

Naveen P R	Team Leader & Core Dev
Rashmi S P	Developer
Sneha H S	Developer
Soujanya M	Developer
Deepa G R	Developer
Kavana N	Developer
Vanashree R	Developer
Deeksha D R	Developer
Saud Shaik	Developer

📋 Phase 1 — Project Planning & Requirements

TASK	OWNER
Project scope definition and requirement gathering	Naveen P R
Technology stack selection (React, Spring Boot, MySQL)	Naveen P R
System architecture design	Naveen P R
Database schema design	Rashmi S P
Task allocation to team members	Naveen P R

🗄️ Phase 2 — Database Design & Setup

TASK	OWNER
ER diagram design	Rashmi S P
MySQL schema creation (all 10 tables)	Sneha H S
Foreign key relationships and constraints	Rashmi S P

⚙️ Phase 3 — Backend Development

TASK	OWNER
Spring Boot project setup and configuration	Naveen P R
Entity classes (Student, Teacher, Admin, Course, etc.)	Rashmi S P
Repository layer (JPA repositories)	Sneha H S
AuthService — login, register, change password	Naveen P R
OtpService — generate, send, validate OTP	Naveen P R
EmailService — Gmail SMTP integration	Naveen P R
AuthController — login/register endpoints	Naveen P R
Student entity and repository design	Rashmi S P
Teacher entity and repository design	Rashmi S P
StudentController — CRUD endpoints	Sneha H S
TeacherController — CRUD endpoints	Soujanya M
CourseController — CRUD endpoints	Deepa G R
AttendanceController — mark and fetch attendance	Soujanya M
MarksController — enter and fetch marks	Kavana N
AssignmentController — CRUD endpoints	Vanashree R
SubmissionController — submission endpoints	Deeksha D R
CourseMaterialController — upload/fetch materials	Saud Shaik
OtpController — forgot password flow	Naveen P R
PasswordResetController — reset endpoints	Naveen P R
SystemSettingsController — maintenance mode	Naveen P R
BulkImportController — CSV upload	Soujanya M
Marks entity and repository design	Sneha H S
Attendance entity and repository design	Sneha H S
SecurityConfig — CORS and Spring Security	Naveen P R

TASK	OWNER
AdminSeeder — default admin auto-creation	Naveen P R

🎨 Phase 4 — Frontend Development

TASK	OWNER
React project setup and folder structure	Naveen P R
Theme system (dark/light mode)	Naveen P R
Landing Page	Kavana N
Login Page (role selector, validation)	Naveen P R
Signup Page	Vanashree R
Forgot Password (3-step OTP flow)	Naveen P R
Admin Dashboard	Naveen P R
Admin — Student Management (CRUD)	Rashmi S P
Admin — Bulk CSV import UI	Rashmi S P
Admin — Teacher Management (CRUD)	Sneha H S
Student profile settings page	Sneha H S
Admin — Course Management	Deepa G R
Admin — System Settings	Naveen P R
Student Dashboard	Naveen P R
Student — Attendance view	Soujanya M
Student — Marks and CGPA view	Kavana N
Student — Assignments view	Vanashree R
Student — Course Materials	Deeksha D R
Teacher Dashboard	Naveen P R
Teacher — Mark Attendance	Soujanya M
Teacher — Enter Marks	Rashmi S P

TASK	OWNER
Teacher — Manage Assignments	Vanashree R
Teacher — Upload Materials	Saud Shaik
Settings Page (profile, password, system)	Naveen P R
Bulk Import (CSV upload UI)	Soujanya M

🔗 Phase 5 — Integration & Bug Fixes

TASK	OWNER
Frontend-backend API integration	Naveen P R
CORS configuration for production	Naveen P R
Bug fix — change password endpoint (admin)	Naveen P R
Bug fix — role switching clears fields	Naveen P R
Bug fix — OTP role validation	Naveen P R
Bug fix — edit modal read-only fields	Rashmi S P
API response validation and error handling	Rashmi S P
Bug fix — profile name update in sidebar	Sneha H S
Frontend form validation improvements	Sneha H S
Bug fix — input focus loss in forgot password	Naveen P R
Bug fix — admin edit student/teacher	Soujanya M

🧪 Phase 6 — Testing

TASK	OWNER
JUnit unit tests — AuthService	Naveen P R
JUnit unit tests — OtpService	Naveen P R
JUnit controller tests — AuthController	Naveen P R

TASK	OWNER
Playwright E2E — Authentication tests	Naveen P R
Playwright E2E — Admin student management	Naveen P R
Playwright E2E — Admin teacher management	Naveen P R
Playwright E2E — Settings and password	Naveen P R
Playwright E2E — Forgot password flow	Naveen P R
Playwright E2E — Student portal	Naveen P R
Playwright E2E — Backend API tests	Naveen P R
Postman collection (26 requests)	Naveen P R

🚀 Phase 7 — Deployment

TASK	OWNER
Dockerfile creation for Spring Boot	Naveen P R
Backend deployment on Render	Naveen P R
Database deployment on Railway (MySQL)	Naveen P R
Frontend deployment on Vercel	Naveen P R
Environment variables configuration	Naveen P R
Production CORS fix	Naveen P R
GitHub repository setup and push	Naveen P R